

Nimbus-V2

Preliminary Design Review

Presented To:

University of California San Diego

Department of Mechanical and Aerospace Engineering

MAE 155B

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Group 2:

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Nimbus-V2 Mission and Scoring Function

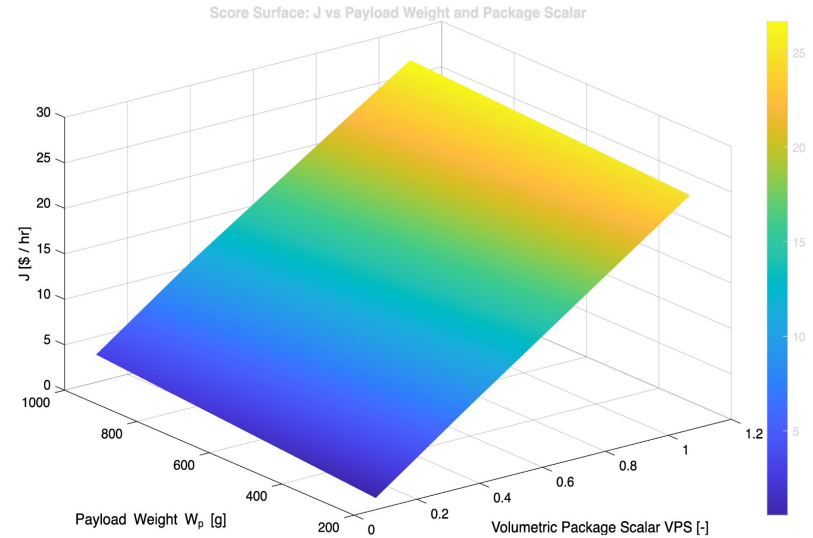
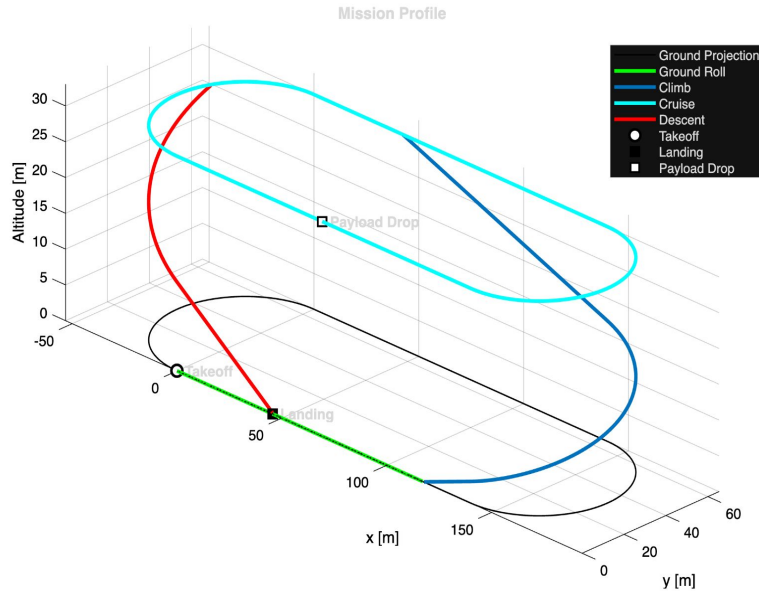


Table 2: Configuration Selection

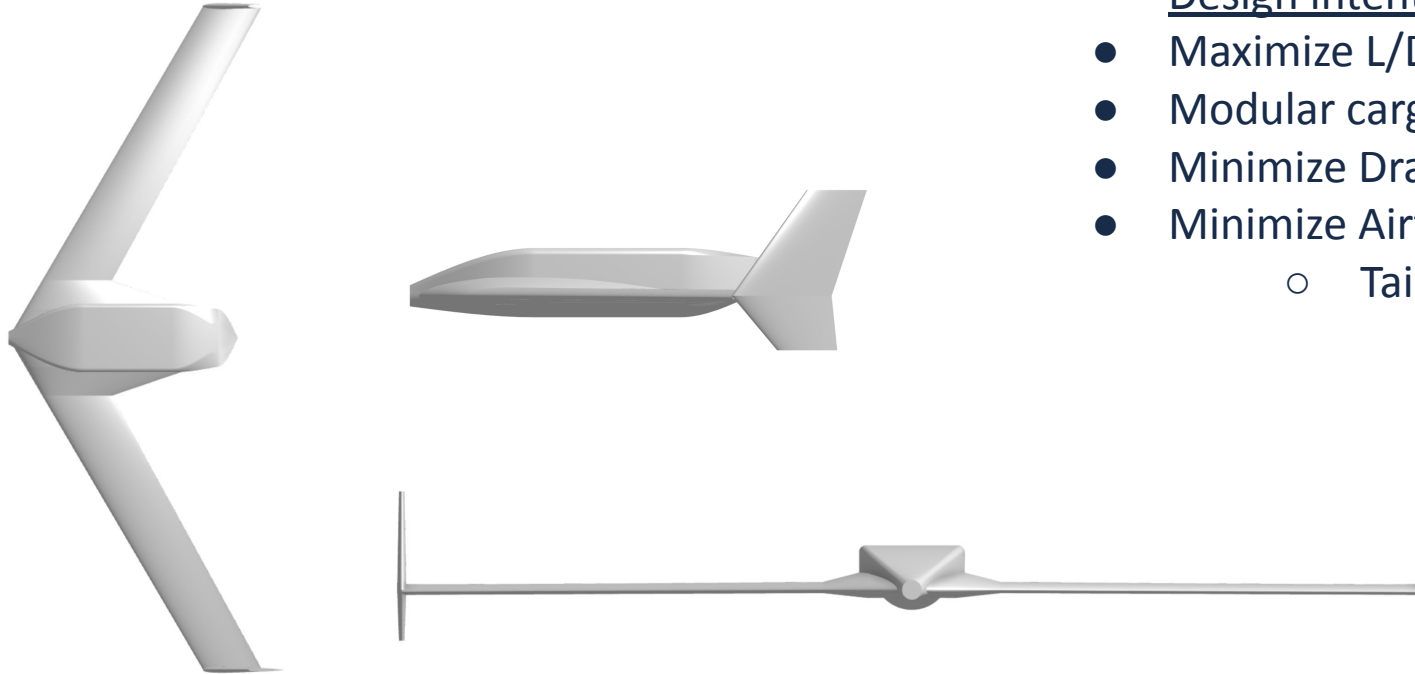
Criterion	Conventional	Flying Wing
Aerodynamic efficiency	-	++
Structural efficiency	-	+
Payload volume efficiency	-	+
Energy efficiency	-	++
Static stability	++	-
Control complexity	++	-
Manufacturing simplicity	+	-
Total	1	3

Criteria:

- Can maximize $J(x)$ with
 - Weight efficiency
 - Aerodynamic efficiency
- Flexible approach to score

Nimbus V2: A Hybrid Approach to Cargo Flying Wings

John

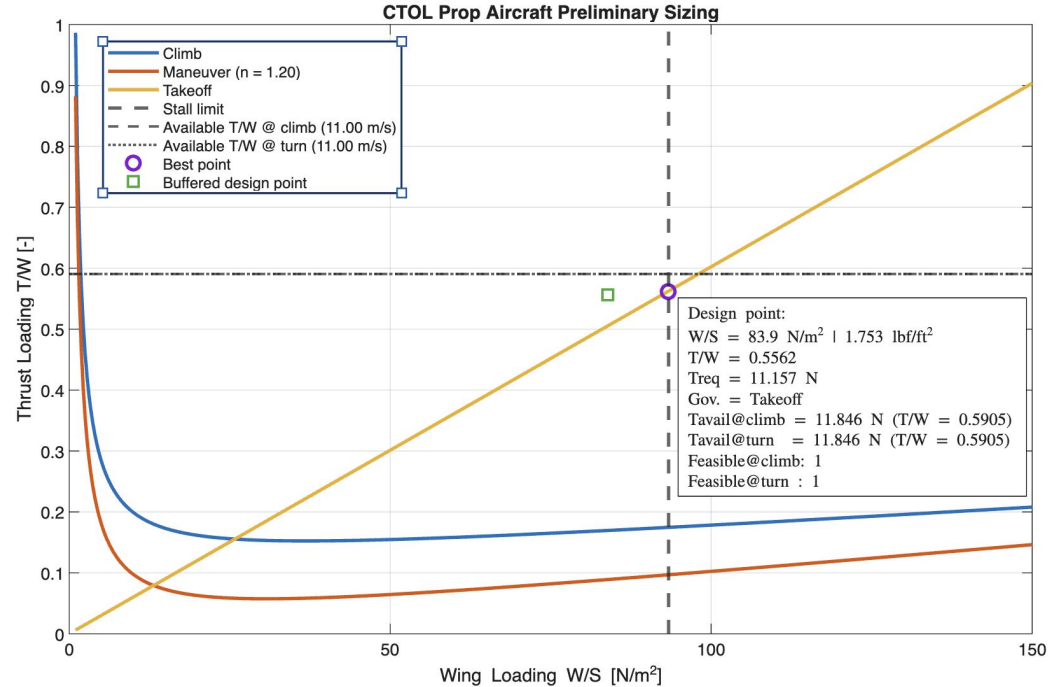


Design intentions:

- Maximize L/D
- Modular cargo approach
- Minimize Drag
- Minimize Airframe Weight
 - Tailless stability

Wing Loading: 83.9 N/m^2

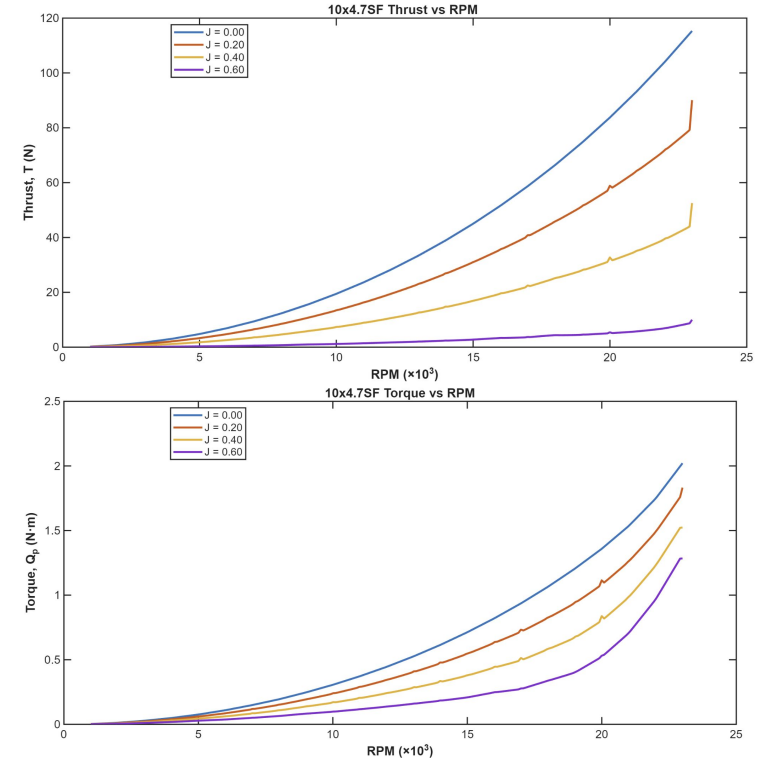
Thrust Loading: 0.5562



Propeller Selection – APC 10×4.7

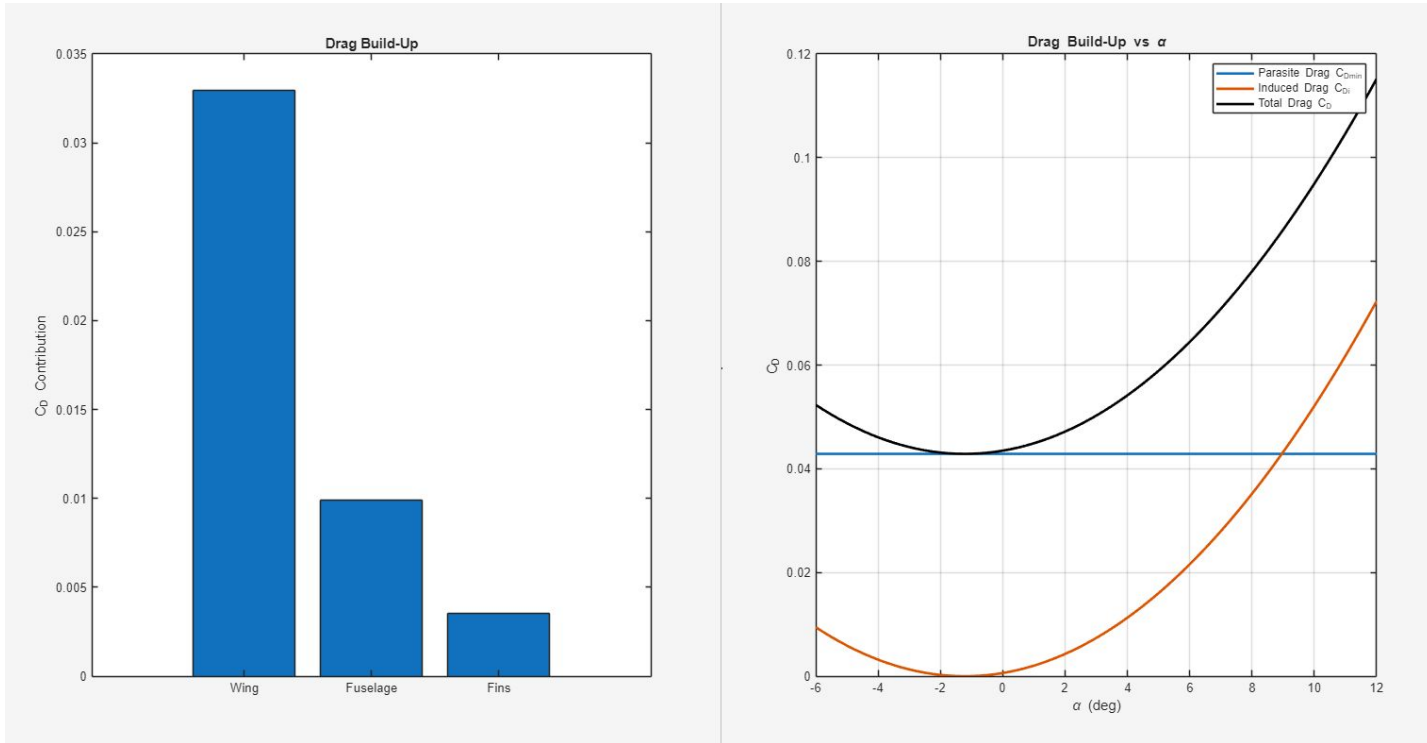
Angel

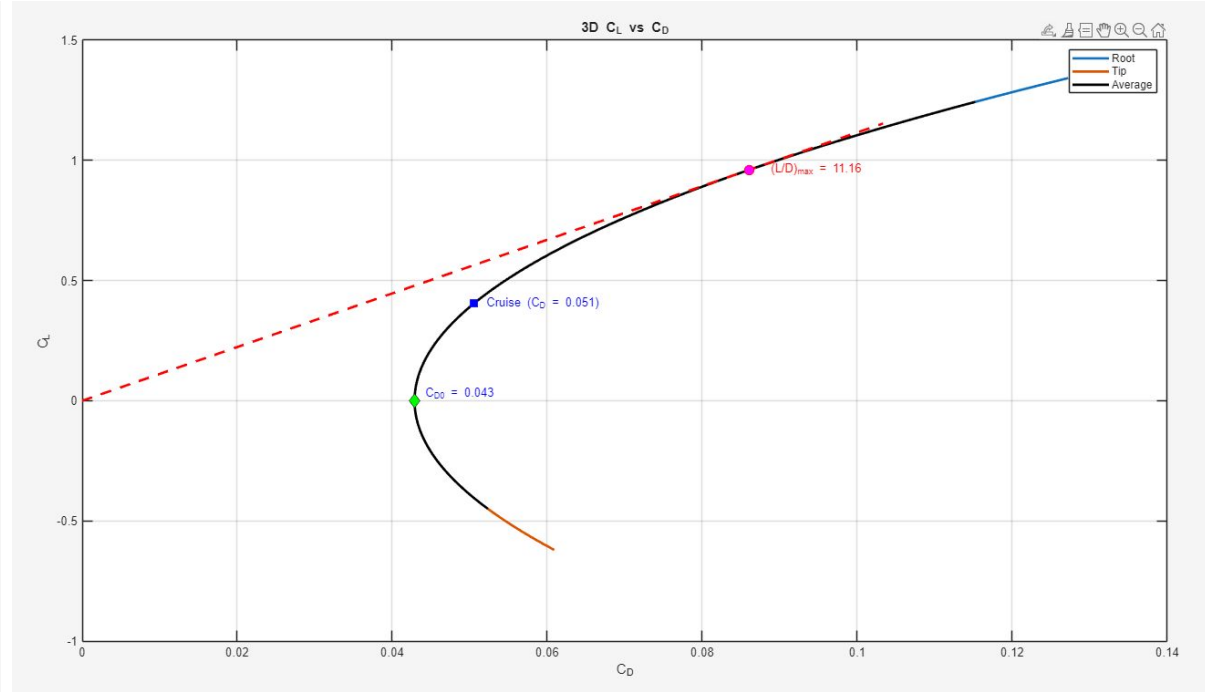
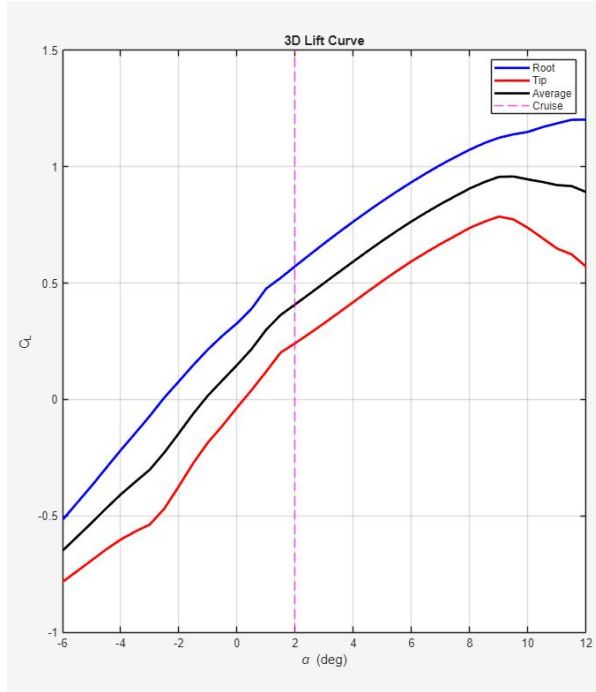
- APC 10x4.7SF
 - Take Off:
 - Thrust = 17.8 N
 - Current = 33.1 A
 - Climb:
 - Thrust = 13.5 N
 - Current = 28.3 A
 - Cruise:
 - Thrust = 8.97 N
 - Current = 23.1 A



Drag Build Up

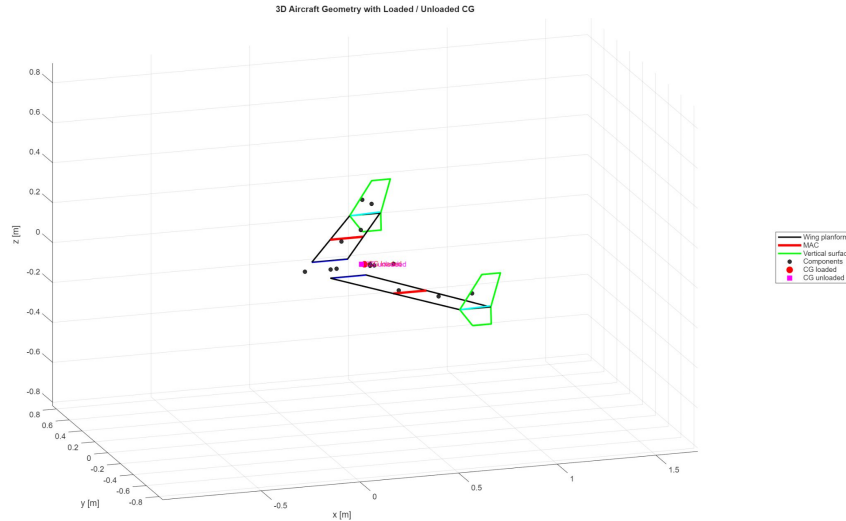
Analisa



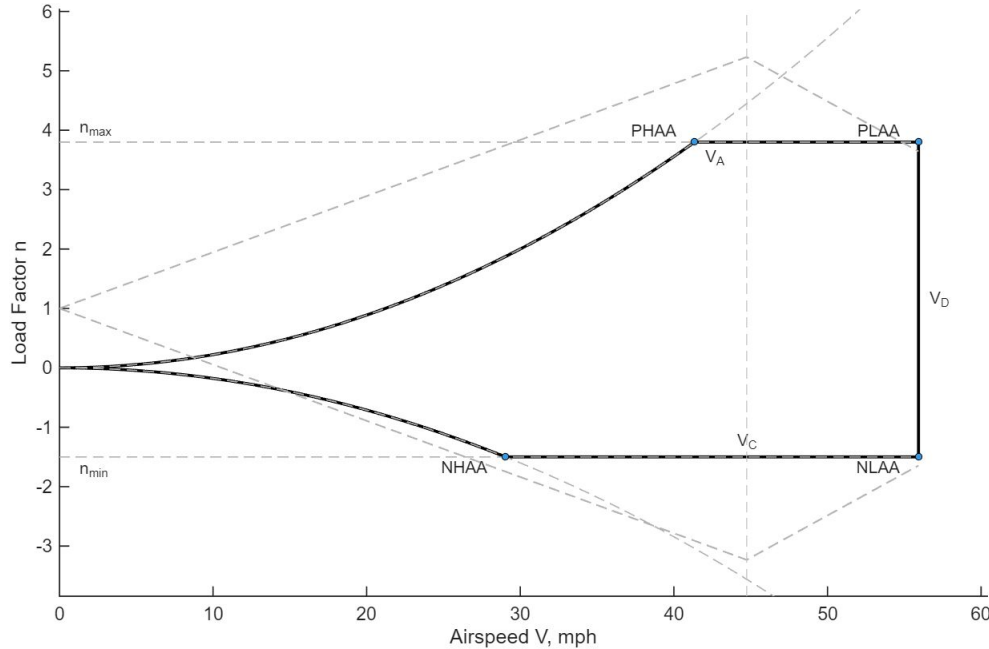


Static Stability Wing Twist Approach

Sara

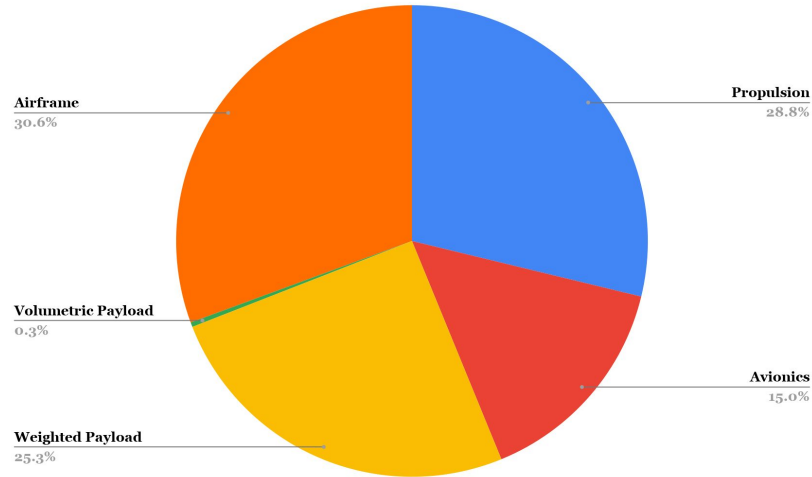


- Wings tips that act as horizontal stabilizers
 - Total twist = -5.9 deg
 - Aerodynamic twist = -3.9 deg
 - Geometric twist = -2.0 deg
- CG = 8.5 % MAC
- Static margin = 16.5 %



- Maneuver and Gust Load Envelope
- Structural Limits
 - -1.5 g to 3.8 g
- Key Speeds
 - $V_c = 44.7$ mph
 - $V_a = 43.6$ mph
 - $V_d = 55.9$ mph

Weight and Structure



- Conservative Airframe estimate
 - Airframe weight as a function of package volume
- Confirms weight efficient design philosophy

Lightweight Hybrid Structure

- Fuselage:
 - Nylon-6 reinforced with glass fiber (PA6-GF)
- Wings:
 - Foam
 - Balsa Wood
- Aerodynamic loads supported using a carbon fiber spar

- Flying wing with twin vertical stabilizers, profit-function optimized
- Gross weight ~2045 g, 800 g payload, 20 m/s cruise
- 17.8 N static thrust vs. 12.1 N required-adequate margin
- Static margin within 10–20% MAC target
- Baseline score: $J = 1.70$ \$/hr -payload volume and L/D are primary optimization levers

- Dynamic Stability
- Stability and aero optimization
- Structures optimization

Questions?